Xinwei Ju's CV

CONTACT

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EDUCATION

University College London, Hawkes Institute 12/2022-12/2027
PhD in Computer Science
Studentship Sponsor: UCL Computer Science Departmental Funds 504450

Imperial College London, Hamlyn Centre 10/2021-10/2022
MRes in Medical Robotics and Image Guided Intervention
Thesis: *Menuai: Restaurant food recommendation system via a transformer-based deep learning model.*
Core Courses: Image Guided Intervention; Medical and Surgical Imaging; Minimally Invasive Surgery; Sensing, Perception and Neuroergonomics

Nanjing Normal University 9/2017- 6/2021
BEng in Electrical Engineering
Core Courses: Analog and Digital Electronics; Electrical Design

PUBLICATIONS

KFS^{4}LAM: Adaptive KeyFrame Selection with Structure-Aware Mapping for Endoscopic Gaussian Splatting SLAM.
Xinwei Ju, Zhehua Mao, Tianyi Song, Danail Stoyanov, Sophia Bano, and Francisco Vasconcelos. *Under review*. [Project Page]

Track2Map: Online Deformable SLAM with Motion-Aware Pose Optimization in Robotic Surgery.
Tianyi Song, Sierra Bonilla, **Xinwei Ju**, Evangelos Mazomenos, Danail Stoyanov, Adam Schmidt, Omid Mohareri, Sophia Bano, and Francisco Vasconcelos. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* (2026). [Paper] [Project Page]

Multi-Modal Monocular Endoscopic Depth and Pose Estimation with Edge-Guided Self-Supervision.
Xinwei Ju, Rema Daher, Danail Stoyanov, Sophia Bano, Francisco Vasconcelos. International Journal of Computer Assisted Radiology and Surgery, 21, 1207–1215 (2026). [Paper] [Project Page]

SegCol Challenge: Semantic Segmentation for Tools and Fold Edges in Colonoscopy Data.
Xinwei Ju*, Rema Daher*, Razvan Caramalau*, Baoru Huang, Negin Ghamsarian, Shunsuke Kikuchi, Atsushi Kouno, Hiroki Matsuzaki, Danail Stoyanov, and Francisco Vasconcelos. *arXiv preprint arXiv:2412.16078* (2024). [Paper] [Project Page]

Mismatched: Evaluating the Limits of Image Matching Approaches and Benchmarks. Sierra Bonilla*, Chiara Di Vece*, Rema Daher*, **Xinwei Ju***, et al. *ECCV Workshop on Map-free Visual Relocalization* (2024). [Paper]

EVEN: An Event-based Framework for Monocular Depth Estimation at Adverse Night Conditions. Peilun Shi, Jiachuan Peng, Jianing Qiu, **Xinwei Ju**, et al. *IEEE ROBIO* (2023). [Paper]

MenuAI: Restaurant Food Recommendation System via a Transformer-based Deep Learning Model. **Xinwei Ju**, Frank Lo, Jianing Qiu, et al. *arXiv preprint* (2022). [Paper]

Clustering Egocentric Images in Passive Dietary Monitoring with Self-supervised Learning. Jiachuan Peng, Peilun Shi, Jianing Qiu, **Xinwei Ju**, et al. *IEEE BHI* (2022). [Paper]

SKILLS

- Research Software Development: Python, PyTorch, CUDA, Git, Docker, Linux.
- Data Management: dataset curation, preprocessing pipelines, storage management.
- Computing Infrastructure: HPC cluster usage, GPU server administration.

WORK EXPERIENCE

Postgraduate Teaching Assistant 10/2023-Present
Department of Computer Science, UCL
- COMP0128 Robotic Control Theory and Systems,
  COMP0137 Machine Vision,
  COMP0233 Research Software Engineering with Python.

Digital Research Technical Champion 11/2025-Present
STEP-UP Programme by UCL, IC and KCL
- Conduct research engineering surveys, workshops, tutorials in department.

Postgraduate Teaching Assistant 11/2023-5/2025
Academic Communication Centre, UCL
- 1-on-1 academic writing tutorials with undergraduate students.
- Assisted staff in organizing academic writing workshops.

ACTIVITIES & TRAINING

Science of Surgery 04/2025
Hawkes Institute, UCL
- Team lead of the Surgical Robotics Vision group during the annual Open Day.
- Designed and delivered accessible demonstrations using interactive games and software to introduce the public to surgical science and our research applications.

Research Software Engineering (RSE) Summer Workshop 06/2025
Centre for Advanced Research Computing, UCL
- Completed practices in research software engineering.
- Strengthened collaborative coding skills in reproducible workflows, data management, version control, testing, and sustainable software development.

Certified Carpentries Instructor 09/2026
The Carpentries
- Certified instructor trained in evidence-based teaching and delivery of hands-on programming and research computing workshops.
- Teach reproducible workflows, version control, testing, and sustainable software development, translating technical concepts for learners with diverse backgrounds.